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Proteome-wide analysis identifies plasma immune regulators of amyloid-beta progression

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A B S T R A C T

While immune function is known to play a mechanistic role in Alzheimer's disease (AD), whether immune proteins in peripheral circulation influence the rate of amyloid-β (Aβ) progression – a central feature of AD – remains unknown. In the Baltimore Longitudinal Study of Aging, we quantified 942 immunological proteins in plasma and identified 32 (including CAT [catalase], CD36 [CD36 antigen], and KRT19 [keratin 19]) associated with rates of cortical Aβ accumulation measured with positron emission tomography (PET). Longitudinal changes in a subset of candidate proteins also predicted Aβ progression, and the mid- to late-life (20-year) trajectory of one protein, CAT, was associated with late-life Aβ-positive status in the Atherosclerosis Risk in Communities (ARIC) study. Genetic variation that influenced plasma levels of CAT, CD36 and KRT19 predicted rates of Aβ accumulation, including causal relationships with Aβ PET levels identified with two-sample Mendelian

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randomization. In addition to associations with tau PET and plasma AD biomarker changes, as well as expression patterns in human microglia subtypes and neurovascular cells in AD brain tissue, we showed that 31 % of candidate proteins were related to mid-life (20-year) or late-life (8-year) dementia risk in ARIC. Our findings reveal plasma proteins associated with longitudinal Aβ accumulation, and identify specific peripheral immune mediators that may contribute to the progression of AD pathophysiology.

1. Introduction

Alzheimer's disease (AD) is characterized by the accumulation of cortical amyloid-beta (Aβ), but emerging data indicate that immune mechanisms can mediate the rate of this heterogeneous disease process. Positron emission tomography (PET) studies using ^{11}C-Pittsburgh compound-B (PiB), a gold-standard *in vivo* measure of cortical Aβ, along with radiotracers of neuroimmune cell types (i.e., microglia, astrocytes) have shown that immune activation across different disease stages can modify trajectories of Aβ accumulation (Rodriguez-Vieitez et al., 2016; Fan et al., 2017). Intrathecal levels of soluble TREM2, a key regulator of the microglial response to Aβ, are associated with differing rates of Aβ progression in pre-symptomatic and symptomatic carriers of familial AD mutations as measured by CSF $Aβ_{42}$ and PiB, respectively (Morenas-Rodríguez et al., 2022). Results from AD mouse models have provided additional support to these findings (Rodriguez-Vieitez et al., 2015; Lee et al., 2021), further suggesting that rates of cortical Aβ change may be dependent on specific immune mediators.

Plasma immune proteins may be sensitive predictors of cortical Aβ progression due to their interactions with target cells in central nervous system (CNS) and the blood–brain-barrier (BBB) (Walker et al., 2019). Among individuals with high Aβ burden (Aβ-positive; Aβ+) but not low Aβ burden (Aβ-negative; Aβ−), elevated plasma levels of IL-6 and CRP are associated with accelerated cortical Aβ accumulation (despite these markers being unrelated to baseline Aβ levels (Oberlin et al., 2021)), while higher plasma levels of IL-12 (a mediator of Th1-macrophage activation) are associated with preserved longitudinal cognition (Yang et al., 2022). Intrathecal studies have similarly reported that the beneficial or deleterious effects linked to certain inflammatory proteins are conditional on disease stage (Pereira et al., 2022; Brosseron et al., 2022), suggesting a dynamic, context-specific relationship between circulating immune proteins and Aβ accumulation that warrants further elucidation. Identification of plasma proteins linked to rates of cortical Aβ change measured *in vivo* may advance our understanding of the peripheral immune mechanisms (and immunologically relevant molecular mediators) associated with the progression of AD pathology, and thereby guide the development of novel, protein-targeting therapeutics. Additionally, such efforts may reveal cost-effective, scalable and clinically relevant biomarkers that could be leveraged to assess future disease risk and screen or track participants for disease-modifying therapies.

Using the Baltimore Longitudinal Study of Aging (BLSA), we measured the levels of 942 immune-related proteins in plasma and identified 32 that were associated with subsequent rates of change in cortical Aβ as measured by PiB PET, including a subset of proteins whose longitudinal trajectories also predicted rates of Aβ accumulation. After examining protein level changes during a multi-decade interval leading up to Aβ onset in the Atherosclerosis Risk in Communities (ARIC) study, we identified genetic variants of CD36, KRT19, and CAT that influenced protein abundance in plasma, predicted rates of cortical Aβ change, and suggested causal relationships with Aβ PET levels using two-sample Mendelian randomization (MR), supporting a potential mechanistic role of these proteins in Aβ accumulation. A subset of the peripheral immune proteins linked to Aβ progression was also associated with tau PET and longitudinal rates of change in plasma biomarkers of AD pathology ($Aβ_{42/40}$, pTau-181), neuronal injury (NfL), and reactive astrogliosis (GFAP). Leveraging ARIC data, we showed that 31 % of these proteins – particularly those linked to slower Aβ progression – were associated with future dementia risk when measured during mid-life or late-life. Finally, we assessed the biological implications and functional relevance of candidate proteins, including their expression patterns in human microglia subtypes, neurovascular cell types and AD brain tissue.

2. Results

2.1. Plasma immune proteins are associated with cortical Aβ accumulation

Using data from the BLSA along with the SomaScan v4.1 Inflammation and Immune Response Panel, we examined how the baseline levels of 942 immune-related proteins in plasma related to subsequent rates of change in cortical Aβ levels (Fig. 1; sTable 1–3; sFig. 1). Participants with at least one Aβ PET scan (n = 196; age = 75.7 [SD = 8.2]; 53.1 % female; 79.6 % white) were eligible for analyses, including 138 with two or more scans (528 total scans; range: 2 to 12) from whom participant-specific standardized rates of change were calculated using linear mixed-effects regression models. Because PiB PET in the presence of low or absent Aβ (i.e., Aβ− participants) can reflect tracer kinetics rather than Aβ deposition (Sojkova et al., 2015), we examined how plasma immune proteins related to rates of Aβ change separately across Aβ+ participants, Aβ− participants, and the total sample, but prioritized detecting immune proteins linked to Aβ progression in Aβ+ participants. The average follow-up time was 5.0 years (median: 4.3, IQR: 2.5, 6.3) with an average of 3.8 (SD = 1.9) scans.

Rates of change in cortical Aβ were elevated for Aβ+ (n = 44) compared to Aβ− (n = 94) participants (Fig. 2A; sFig. 2), which equated to an additional increase of 4.38 centiloids/year. Of the 942 immune-related plasma proteins examined, 32 were associated with subsequent rates of Aβ accumulation among Aβ+ participants at an uncorrected *p*-value of 0.05 in models adjusted for demographic, physiological and comorbid variables (henceforth referred to as *candidate proteins*; Fig. 2B; sTable 4). An anticoagulant/tumor-suppressant (TFPI2), a lipoprotein (APOE2), and a multifunctional gene regulator (NPM1) were the top plasma proteins associated with greater Aβ accumulation, whereas a ubiquitin–proteasome regulator (CUL1), a polyamine enzyme (SMS), and a T-cell regulator (FGL2) demonstrated the strongest associations with lower rates of Aβ accumulation. Although total APOE, APOE3, and APOE4 were associated with significantly greater Aβ accumulation, the surprising positive relationship with APOE2 (i.e., inconsistent with the protective effect of APOE2 genotype) may be attributed to conserved epitopes among APOE gene products, and the corresponding limited selectivity of these aptamers for different APOE isoforms.

Due to this study's exploratory nature, the size of the proteomic panel, and modest number of Aβ+ participants, we had limited power to detect significant effects at an FDR-corrected threshold and therefore did not adjust p values for multiple comparisons. While QQ plots (i.e., observed p values compared to expected p values from a theoretical $\chi 2$ distribution) did not suggest inflated Type-I error ($\lambda = 0.68$; sFig. 4A), permutation testing (100 rounds with Aβ slopes randomly shuffled across participants) did highlight the potential for the identification of false positives. These conflicting patterns may be attributed to the inherently interrelated structure of the immune proteome (i.e., observations are not truly independent) or other unknown sources of bias (sFig. 4B). Notably, the 32 candidate proteins' p values obtained from permutation testing were rarely (if at all) lower than their observed p values in our primary analyses, suggesting that many of the detected associations are not due to random chance (sTable 5).

Notably, only one of the 32 candidate proteins (CD36) was associated with Aβ status (+/−) at baseline, suggesting limited overlap between the peripheral immune proteins associated with rates of Aβ accumulation and those associated with elevated Aβ (Fig. 2C; sTable 6). In addition to steeper rates of Aβ progression and increased odds of Aβ+, higher CD36 was linked to a younger estimated age of Aβ+ onset (Fig. 2D; sTable 7). Candidate proteins showed stronger associations with Aβ progression (average $\log_2$ beta coefficient: 0.839, range: 1.22–1.57) than canonical AD and related dementia (ADRD) plasma biomarkers, including $A\beta_{42/40}$, pTau-181 and pTau-231 (average $\log_2$ beta coefficient: 0.099 range: 0.01–0.21) (Fig. 2E). Plasma levels of one candidate protein (APOE2) were associated with greater Aβ accumulation across Aβ+ participants, Aβ− participants, and the whole sample, whereas another candidate protein (CAT) was associated with different effects on Aβ progression depending on whether participants were Aβ+ or Aβ− (Fig. 2E; sFig. 3A-D). Proteins associated with more rapid Aβ accumulation among Aβ+ participants tended to be associated with attenuated Aβ accumulation among Aβ− participants (sFig. 3C). With few exceptions (e.g., CDCP1, CCL2), results were consistent in models adjusted for physiological and comorbid variables only (sTable 4). In sensitivity analyses adjusting for sample storage time, effect estimates remained largely similar, with the exception of APOE2, COL5A1, KITLG, SH2D1A, SMS, which showed effect size decrements > 10 % (sTable 4). Candidate protein associations did not show effect modification by sex, suggesting their relationships with Aβ accumulation did not statistically differ between men and women (sTable 4).

To understand how the trajectory of candidate protein levels related to the rate of cortical Aβ progression, we calculated rates of change in protein levels from participants who had two or more blood draws and PET scans (n = 133; 339 total blood samples; average follow up: 5.7 years, 2.6 blood samples; range: 2 to 6). Consistent with the effects of baseline protein levels, greater increases in two proteins (S100A8, NCF1) and greater reductions in another two (KRT19, APOC3) were associated with blunted Aβ accumulation among Aβ+ participants, while greater reductions in another two (LTA, CSF3) were associated with higher rates of Aβ accumulation (Fig. 2F-G; sTable 8–9). Although higher levels of baseline SH2D1A and FGL2 predicted higher and lower rates of Aβ progression, respectively, greater longitudinal increases in these proteins were associated with lower and higher rates of Aβ progression, respectively. While their levels did not change significantly over time, rates of change in several other candidate proteins, including APOE3, IRF4, CAT, NPM1, were also linked to altered Aβ accumulation.

For the twelve candidate proteins whose baseline levels and rates of change were related to altered Aβ accumulation in BLSA, we utilized data collected in the ARIC study (n = 272; age = 75.7 [SD = 5.2]; 56.2 % female; 59.6 % white; sTable 10; sFig. 5) to determine if preceding rates of change in these protein levels might be associated with subsequent Aβ status. Proteins were measured in late-life at the time of Aβ PET scan, as well as in mid-life, 20 years prior. Of the candidate proteins measured in ARIC, greater increases in one (CAT) over mid- to late-life were related to increased odds of Aβ+ (sFig. 6; sTable 11).

2.2. *Candidate protein genetic variation predicts cortical Aβ accumulation and Aβ levels*

To gain mechanistic insights, we then examined whether genetic variants that influence candidate protein levels in plasma were also related to rates of cortical Aβ change using protein quantitative trait loci (pQTLs) identified by the deCODE study (n = 35,559) (Ferkingstad et al., 2021). Among BLSA participants with available genotyping data, we replicated associations for 13 of the 21 candidate proteins with available pQTLs (i.e., at least one SNP per protein was associated with protein abundance in plasma; sTable 3, 12; sFig. 7). Consistent with the direction of candidate protein effects, *trans* variants related to increased levels of CD36 (rs977371848; Fig. 3A-B) and KRT19 (rs114115200; Fig. 3C-D) were associated with greater Aβ accumulation. A promoter-specific, *cis* variant linked to decreased levels of CAT (rs1001179; Fig. 3E-F) was also related to lower rates of Aβ accumulation. Although they did not significantly relate to protein levels in the BLSA, previously identified pQTLs for several other candidate proteins (TFPI2, RPSA, KITLG, SMS, ETNK1) were similarly related to altered rates of Aβ progression. Given the role of *APOE* in Aβ deposition and AD risk, we confirmed that no pQTLs of interest were located in the *cis* region of *APOE*.

To formally test if genetic variation that influences CAT, CD36, and KRT19 protein levels in plasma may be causally related to brain amyloidosis, we combined their pQTLs with summary statistics from a

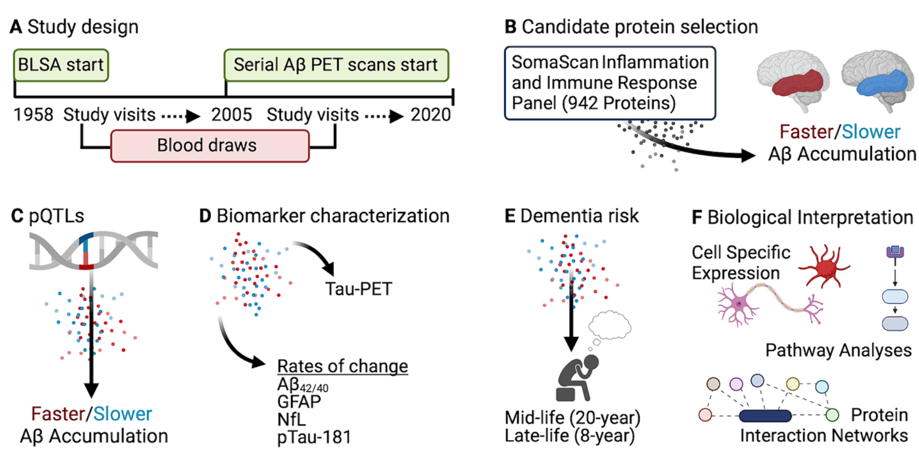

Fig. 1. Study design. A Baltimore Longitudinal Study of Aging (BLSA) participants received comprehensive health and functional screening evaluations (including blood draws) at each study visit (study initiated in 1958). Amyloid beta (Aβ) PET scans for ^{11}C-Pittsburgh compound-B (PiB) were initiated in 2005. Associations of SomaScan proteomic measurements with neurocognitive outcomes used blood samples collected at the baseline visit of each respective outcome (e.g., PET scan). Due to BLSA's continuous enrollment, participants entered the study at different times and varied with respect to follow-up times. B Candidate proteins were selected if their plasma levels were associated with rates of change in cortical Aβ. C Genetic variants that influenced candidate protein levels (plasma protein quantitative trait loci [pQTLs]) were associated with rates of change in cortical Aβ. D Candidate proteins were related to cross-sectional tau-PET levels and longitudinal changes in plasma ADRD biomarkers ($A\beta_{42/40}$, GFAP, NfL, pTau-181). E Candidate proteins measured in mid-life and late-life were related to 20- and 8-year dementia risk, respectively, in the Atherosclerosis Risk in Communities (ARIC) study. F The biological implications and functional relevance of candidate proteins were assessed using a variety of complementary analytic tools and open-source databases. Key: Aβ, amyloid beta; BLSA, Baltimore Longitudinal Study of Aging; PET, positron emission tomography.

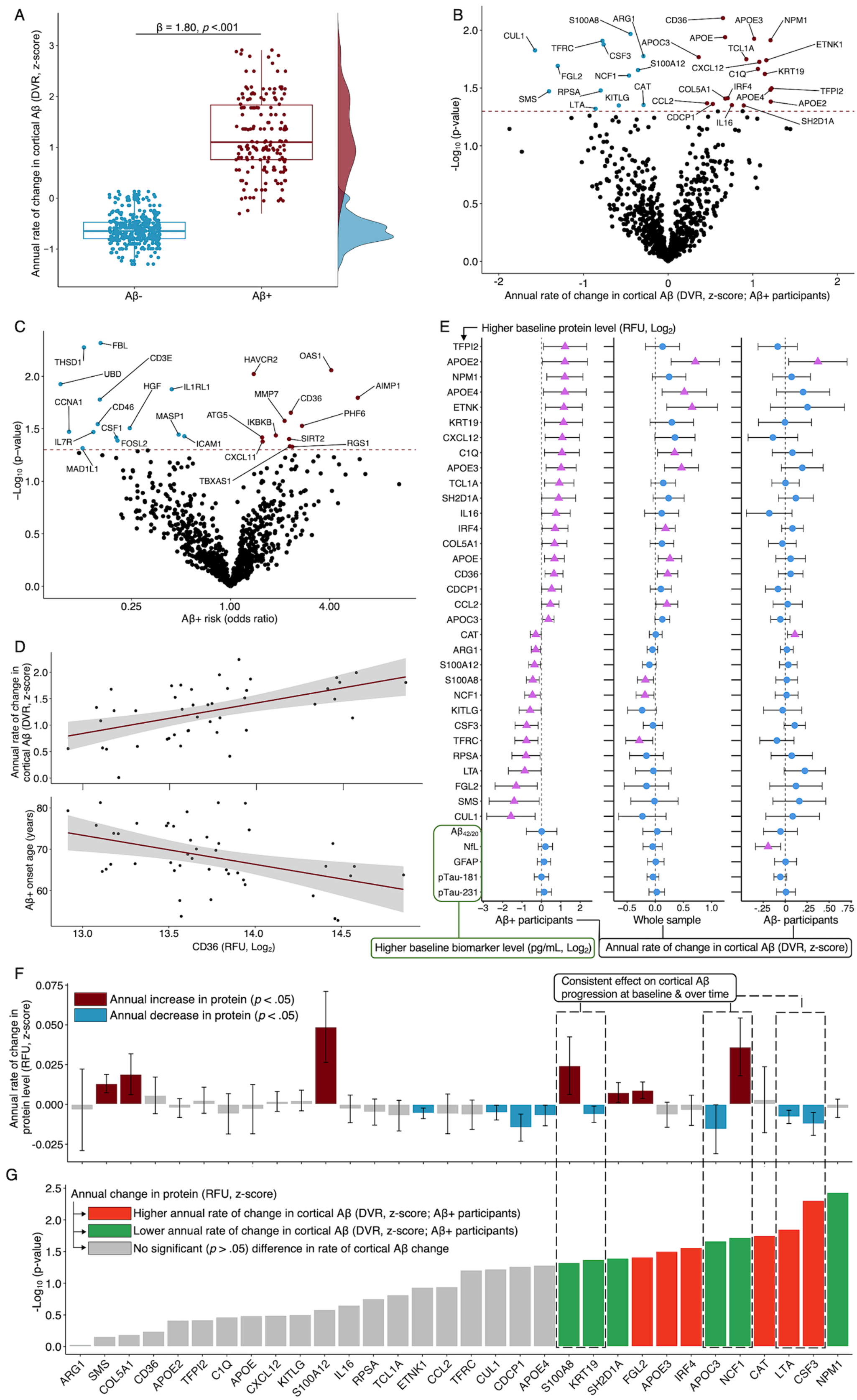

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Fig. 2. Plasma immune proteins associated with cortical Aβ accumulation (candidate proteins) A Boxplot (and corresponding density plots along y-axis) shows the distribution of annual rates of change in mean cortical amyloid beta (Aβ), displayed across amyloid-negative (Aβ−) and amyloid-positive (Aβ+) participants. Participant-specific standardized rates of change were extracted from linear mixed-effects regression models. The beta (β) and corresponding *p*-value reflect differences in rates of Aβ accumulation as a function of Aβ status (+/−); results were derived from linear regression models adjusting for baseline age, sex, race, education, *APOE ε*4, sample batch, and a comorbidity index (i.e., obesity, hypertension, diabetes, cancer, ischemic heart disease, chronic heart failure, chronic kidney disease and chronic obstructive pulmonary disease). B Volcano plot shows the annual rates of change in cortical Aβ associated with baseline protein levels ($\log_2$) among Aβ+ participants. Results were derived from covariate-adjusted linear regression models. Proteins above the dashed line were statistically significant ($p < 0.05$). Red and blue dots indicate proteins associated with higher and lower rates of Aβ accumulation, respectively. C Volcano plot shows the risk of being Aβ+ associated with baseline protein levels ($\log_2$). Odds ratios were derived from covariate-adjusted logistic regression models. Proteins above the dashed line were statistically significant ($p < 0.05$). Red and blue dots indicate proteins associated with increased and decreased Aβ+ risk, respectively. D Scatterplots and lines of best fit show annual rates of change in cortical Aβ and Aβ+ onset age (the estimated age at which a participant became Aβ+) associated with CD36 protein levels ($\log_2$). Results were derived from covariate-adjusted linear regression models. E Forest plots show candidate proteins significantly associated annual rates of change in cortical Aβ, displayed across Aβ+ participants, the whole sample, and Aβ− participants. Results were derived from covariate-adjusted linear regression models; analyses with the whole sample also adjusted for Aβ status (+/−). F Bar graph shows annual rates of change in candidate protein levels ($\log_2$). Participant-specific standardized rates of change were extracted from linear mixed-effects regression models. Error bars reflect 95 % confidence intervals. G Bar graph shows the significance ($-\log_{10}$ *p*-values) of associations between annual rates of change in candidate protein levels and annual rates of change in cortical Aβ among Aβ+ participants. Results were derived from covariate-adjusted linear regression models. Key: Aβ, amyloid beta; DVR, distribution volume ratio; RFU, relative fluorescent units.

GWAS of cortical Aβ PET levels (n = 13,409) (Ali et al., 2023), and conducted two-sample MR (Fig. 3G; sTable 13, 14). We found evidence supporting a causal relationship of plasma CD36 abundance with increased Aβ, and of plasma KRT19 abundance with decreased Aβ. Although we also found evidence suggesting a relationship of plasma CAT abundance with decreased Aβ PET levels, this relationship was not statistically significant (p = 0.074). Together, these findings provide further support for the causal relationship between the abundance of several candidate proteins in plasma and rates of cortical Aβ accumulation.

2.3. Proteins associated with cortical Aβ accumulation predict level and change in AD biomarkers

We then asked how candidate proteins related to cross-sectional tau-PET levels in the BLSA (n = 78; sTable 3; sFig. 8). Several candidate proteins linked to more rapid cortical Aβ accumulation were associated with lower regional tau-PET levels at baseline, including CXCL12, TFIP2 and IRF4 (Braak I-II) as well as NPM1 (Braak III-IV) (Fig. 3H; sTable 15), suggesting that a subset of the candidate proteins may have a pleiotropic association with two hallmarks of AD. Similar patterns were found in minimally adjusted, stratified analyses (Aβ+ n = 20; Aβ− n = 58), where we also noted CAT's association with greater tau-PET levels in the ITG among Aβ+ participants (sFig. 11A).

We also examined whether candidate proteins were related to longitudinal changes in plasma ADRD biomarkers in the BLSA (n = 638 for $A\beta_{42/40}$ ratio, NfL, GFAP; n = 541 for pTau-181) measured over a median follow-up time of 5.1 years (IQR: 4.0, 7.1) for $A\beta_{42/40}$, GFAP and NfL analyses and 5.0 years (IQR: 4.0, 7.0) for pTau181 analyses (sTable 3, sFig. 9, 10). Given that ADRD biomarker trajectories may be influenced by the degree of underlying AD pathology, these analyses adjusted for Aβ status among participants without Aβ PET using plasma Aβ status (defined by plasma $A\beta_{42/40}$ because it was available in a larger set of participants and illustrated comparable performance for discriminating Aβ PET status relative to other biomarkers; sFig. 11B-C). Eight (25 %) candidate proteins linked to greater cortical Aβ accumulation were associated with changes in ADRD biomarkers, including CXCL12 (higher increases in GFAP), C1Q (higher increases in NFL), and ETNK1 (greater decreases in $A\beta_{42/40}$, indicative of greater increases in brain Aβ), while two others (IL16, COL5A1) were related to attenuated increases in pTau-181 (Fig. 3I; sTable 16). Of the proteins associated with lower rates of cortical Aβ progression, one was related to greater increases in GFAP (KITLG), one was associated with lower increases in pTau-181 (CSF3), and another was related to lower increases in NfL (SMS), especially among Aβ+ participants (sFig. 11D).

2.4. Candidate proteins predict mid- and late-life dementia risk

To determine if plasma proteins associated with cortical amyloid accumulation might also influence dementia risk, we utilized data collected in the ARIC study to examine the associations of candidate proteins measured in mid-life (n = 11,596; age = 57.1 [SD = 5.7]; 55.7 % female; 77.0 % white) and late-life (n = 4,288; age = 75.2 [SD = 5.0]; 57.9 % female; 81.8 % white) with 20- and 8-year dementia risk, respectively (sTable 10; sFig. 12). Of the 26 candidate proteins measured in ARIC, eight (31 %) were significantly associated with risk of incident dementia diagnosis. Higher plasma APOE levels, which were related to greater rates of Aβ accumulation in the BLSA, were associated with greater dementia risk in ARIC when measured during mid-life. Surprisingly, higher plasma APOC3 levels, which were also linked to greater rates of Aβ accumulation in the BLSA, were associated with decreased dementia risk in ARIC when measured during late-life (Fig. 3J; sTable 17). Among candidate proteins associated with decreased Aβ progression in the BLSA, four were linked to lower dementia risk in ARIC when measured during mid-life (KITLG, TFRC, SMS, CUL1), while another two were associated with increased dementia risk over that same period (CAT, S100A12). Notably, the S100A12 association with increased dementia risk persisted into late-life, whereas TFRC transitioned from being associated with reduced risk at mid-life to being associated with greater risk during late-life. Although not statistically significant, several other protein-dementia associations (CAT, C1Q, IRF4) showed similar transitions across mid-life and late-life, suggesting that the relationships between some peripheral immune proteins and neurocognitive outcomes may be dependent on disease stage.

2.5. Candidate protein correlations across biofluids

To understand whether and to what extent candidate proteins in peripheral circulation relate to CNS levels of the same protein, we next determined the extent to which protein abundance in plasma correlated with corresponding protein abundance in CSF. Leveraging SomaScan plasma and CSF protein measurements from the Johns Hopkins Neurology Clinic (n = 194), we found the levels of seven proteins (CCL2 [r = 0.21], COL5A1 [r = 0.22], CUL1 [r = 0.15], KITLG [r = 0.14], NCF1 [r = 0.25], CXCL12 [r = 0.19], TFRC [r = 0.16]) were positively correlated across biofluids (sTable 18).

2.6. Cell specific and AD expression patterns

We then leveraged publicly available single-cell RNAseq data to investigate the cell specific expression patterns of cognate genes encoding candidate proteins. Five cognate genes (*CUL1*, *APOE*, *TFRC*, *LTA*, *KITLG*) were highly expressed in at least one CNS cell type, and

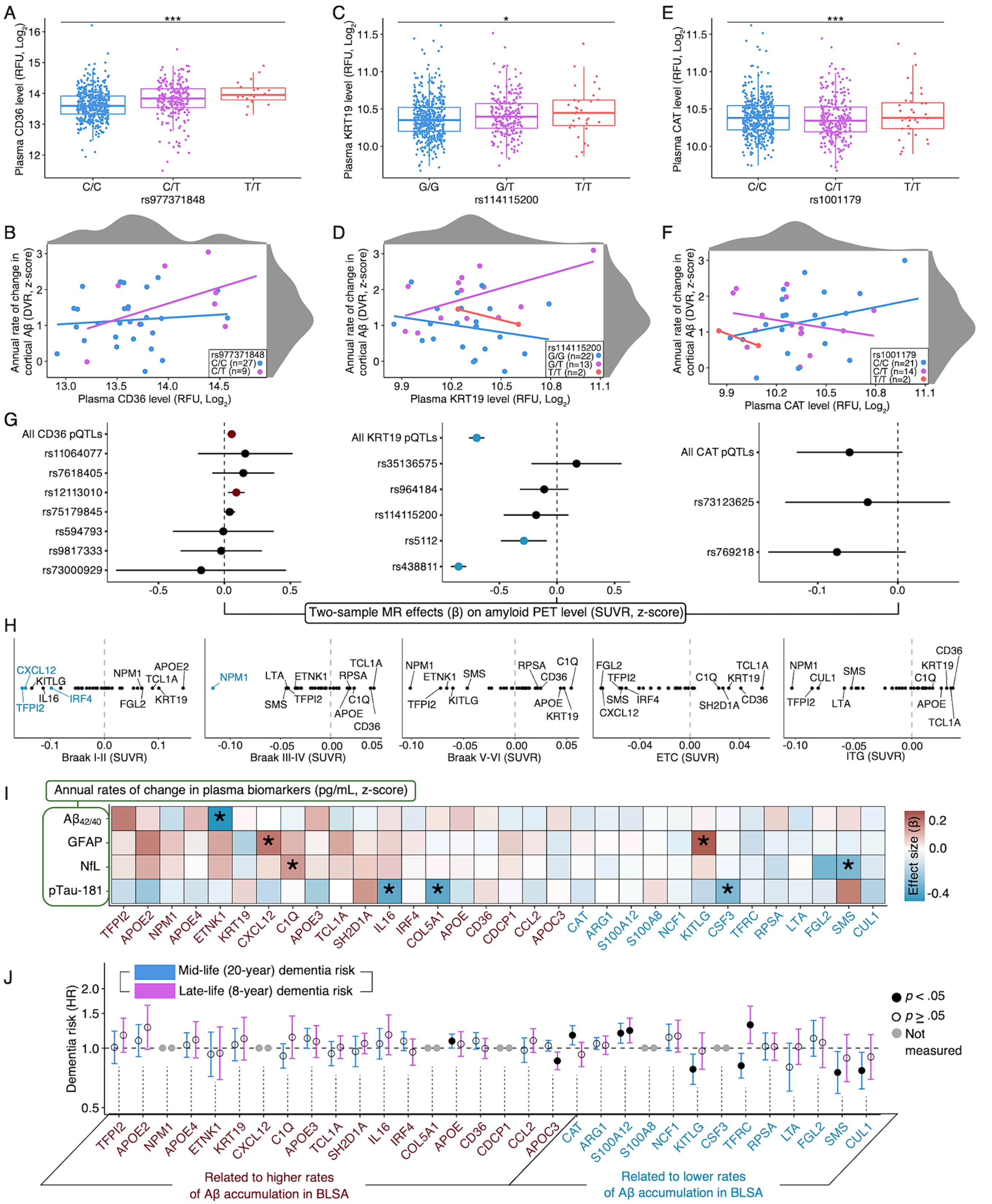

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Fig. 3. Genetic variants that influence plasma levels of candidate proteins predict cortical Aβ accumulation and Aβ levels; candidate proteins are associated with regional tau-PET and ADRD biomarkers, as well as mid- and late-life dementia risk. A, C, E Boxplots show CD36, KRT19, and CAT plasma protein levels ($\log_2$) associated with rs977371848, rs114115200, and rs1001179 genotypes, respectively. B, D, F Scatterplots and lines of best fit show CD36, KRT19, and CAT plasma protein levels ($\log_2$) associated with annual rates of cortical Aβ change displayed across rs977371848, rs114115200, and rs1001179 alleles, respectively. Differences in candidate protein levels associated with a given SNP were derived from multiple linear regression models adjusted for covariates used in the pQTL discovery cohort, namely baseline age and sex. Results examining rates of cortical Aβ change were derived from linear regression models adjusting for baseline age, sex, race, education, *APOEε*4, sample batch, and a comorbidity index (i.e., obesity, hypertension, diabetes, cancer, ischemic heart disease, chronic heart failure, chronic kidney disease and chronic obstructive pulmonary disease). All relationships displayed in A-F were statistically significant ($p < 0.05$). G Forest plots display the results of two-sample Mendelian randomization analyses that assessed the relationships between genetically determined plasma protein levels and Aβ PET levels. Plasma pQTLs were obtained from deCODE Genetics (n = 35,559). Aβ PET level summary statistics were obtained from a Washington University GWAS (n = 13,409). Beta coefficients and 95 % confidence intervals reflect inverse variance weighted (multiple SNPs) or Wald ratio (single SNPs) statistics. Red/blue circles indicate statistically significant associations ($p < 0.05$). H Jitter plots show differences in regional tau-PET levels associated with baseline candidate protein levels ($\log_2$). Results were derived from covariate-adjusted linear regression models. The five proteins with the highest and lowest beta (β) estimates are labeled, with blue or red labels indicating statistically significant ($p < 0.05$) negative and positive associations, respectively. I Heatmap shows the annual rates of change in ADRD biomarkers associated with baseline candidate protein levels ($\log_2$). Participant-specific standardized rates of change were extracted from linear mixed-effects regression models. Results were derived from covariate-adjusted linear regression models, with additional adjustment for eGFR. *Indicates statistically significant associations ($p < 0.05$). J Forest plot shows the associations of candidate proteins with mid-life (20-year) and late-life (8-year) dementia risk in the Atherosclerosis Risk in Communities (ARIC) study. Results were derived from Cox proportional hazards regression models adjusting for age, sex, race-center, education, *APOEε*4, eGFR-creatinine, and cardiovascular risk factors (BMI, diabetes, hypertension, and current smoking status). Key: Aβ, amyloid beta; ETC, entorhinal cortex; DVR, distribution volume ratio; HR, hazard ratio; ITG, inferior temporal gyrus; RFU, relative fluorescent units; SUVR, standardized uptake value ratio.

another ten were undetectable in CNS cells; among all 76 cell types, cognate genes were most frequently enriched in myeloid cells, including Kupffer cells, dendritic cells, macrophages, and monocytes (sTable 19, 20; sFig. 13).

Bulk RNA-seq results from comprehensive post-mortem tissue collections showed that every cognate gene with available expression data was differentially expressed at the RNA level in AD, with another three (CAT, APOE, NPM1) also differentially expressed at the protein level (sTable 21). Single cell RNA-seq showed 18 (75 %) cognate genes were differentially expressed in one or more prominent CNS cell types in AD; here, differential expression was most evident in neurons and astrocytes, and these patterns were more robust with more advanced AD pathology (sTable 22; sFig. 14A-C). Among neurovascular cell types, 21 (81 %) cognate genes were differentially expressed in AD, where patterns of AD-related expression were most apparent in hippocampal astrocytes, oligodendrocytes, vascular smooth muscle cells and meningeal fibroblasts (sTable 23); candidate proteins with potential mechanistic links to Aβ progression showed altered expression most prominently in ependymal cells (CD36), arterial endothelial cells (KRT19), and cortical astrocytes (CAT) (sFig. 14D). To determine if AD-related patterns of cognate gene expression were translatable to preclinical models, we leveraged published results which assessed protein and gene expression in brain tissue of AD transgenic mice and their wild type littermates (Kim et al., 2019; Sierksma et al., 2020; Zhou et al., 2020; Boeddrich et al., 2023) (sTable 24). Of the candidate proteins detected via mass spectrometry, eight (100 %) were differentially expressed in 5XFAD mice. Of the cognate genes detected via bulk RNAseq, 10 (63 %) were differentially expressed in APP/PS1 mice. Of the cognate genes detected via single nuclei RNAseq, 14 (56 %) were differentially expressed in 5XFAD mice, especially in neuronal cell types.

Together, these findings indicate i) the majority of cognate genes are differentially expressed in AD ii) cognate genes of candidate proteins linked to steeper amyloid accumulation tend to be upregulated in AD brain tissue and neurovasculature, iii) cognate gene expression in peripheral immune cells and neurons likely contributes to the associations of candidate proteins with Aβ accumulation, and iv) peripheral immune signaling may affect Aβ progression through interactions with neurovascular and meningeal cell types (Yang et al., 2021; Da Mesquita et al., 2018).

2.7. Expression in human microglia subtypes

Although we did not detect robust expression of cognate genes across total microglia cell populations in preceding analyses, accumulating data indicate that altered expression among different subtypes of microglia cells may nonetheless be a primary determinant of the rate of amyloid progression in AD (Keren-Shaul et al., 2017; Mancuso et al., 2024). To investigate if expression patterns in different microglia populations may be contributing to the associations with cortical amyloid detected among immune proteins in plasma, we used single-cell transcriptomics from human microglia xenotransplanted to mouse brains (138,577 cells, n = 106) to examine the expression of cognate genes encoding candidate proteins across eight distinct microglia subtypes (Mancuso et al., 2024) (i.e., differential expression of each subtype compared to all others), including homeostatic (HM), cytokine responsive-1 and -2 (CRM1, CRM2), transitioning cytokine responsive microglia (tCRM), interferon response (IRM), antigen-presenting (HLA), ribosomal (RM) and disease-associated (DAM). Of the 24 genes detected in one or more microglia subtype, 21 (88 %) were differentially regulated in at least one subtype (Fig. 4A; sTable 25). Altered expression was most frequently observed in HM, where 14 of the 18 differentially expressed genes were downregulated. While expression was also frequently altered in DAM and IRM, two well-characterized phenotypes exhibited in response to Aβ (Deczkowska et al., 2018; Sala Frigerio et al., 2019), differentially expressed genes in these subtypes were primarily upregulated (i.e., 8 of 14 in DAM, and 14 of 17 in IRM). Cognate genes encoding candidate proteins linked to steeper amyloid accumulation in the BLSA tended to be upregulated in DAMs and IRMs, but downregulated among HMs. To confirm DAM enrichment among candidate proteins, we also assessed cognate gene expression in a murine microglia transcriptomes reported elsewhere (Keren-Shaul et al., 2017); of the 12 genes detected in this independent sample, 9 (75 %) were differentially expressed in DAM, further suggesting a robust enrichment of Aβ−sensitive candidate proteins (sTable 26).

2.8. Candidate protein biology

Existing studies externally validated 21 (66 %) candidate protein measurements (SOMAmer reagents) using one or more biochemical techniques (e.g., mass spectrometry, proximity extension assay etc.,) (sTable 27). Pathway analyses showed candidate proteins were most strongly enriched for atherogenic, retinoic acid and macrophage-specific reactive oxygen species (ROS) signaling cascades (Fig. 4B; sTable 28; sFig. 15–17), which have been previously implicated in AD (Duggan et al., 2020; Wingo et al., 2020; Goodman and Pardee, 2003). We also identified pro-inflammatory mediators (TNF, IL1-β, lipopolysaccharide (LPS)) as their strongest upstream regulators, consistent with prior studies showing pro-inflammatory mechanisms underlying neurodegeneration may be mediated through alterations in the plasma immune proteome (Duggan et al., 2023; Zhong et al., 2023)(Fig. 4C; sTable 29). Further examination of their canonical functions revealed candidate proteins may influence rates of Aβ accumulation through a

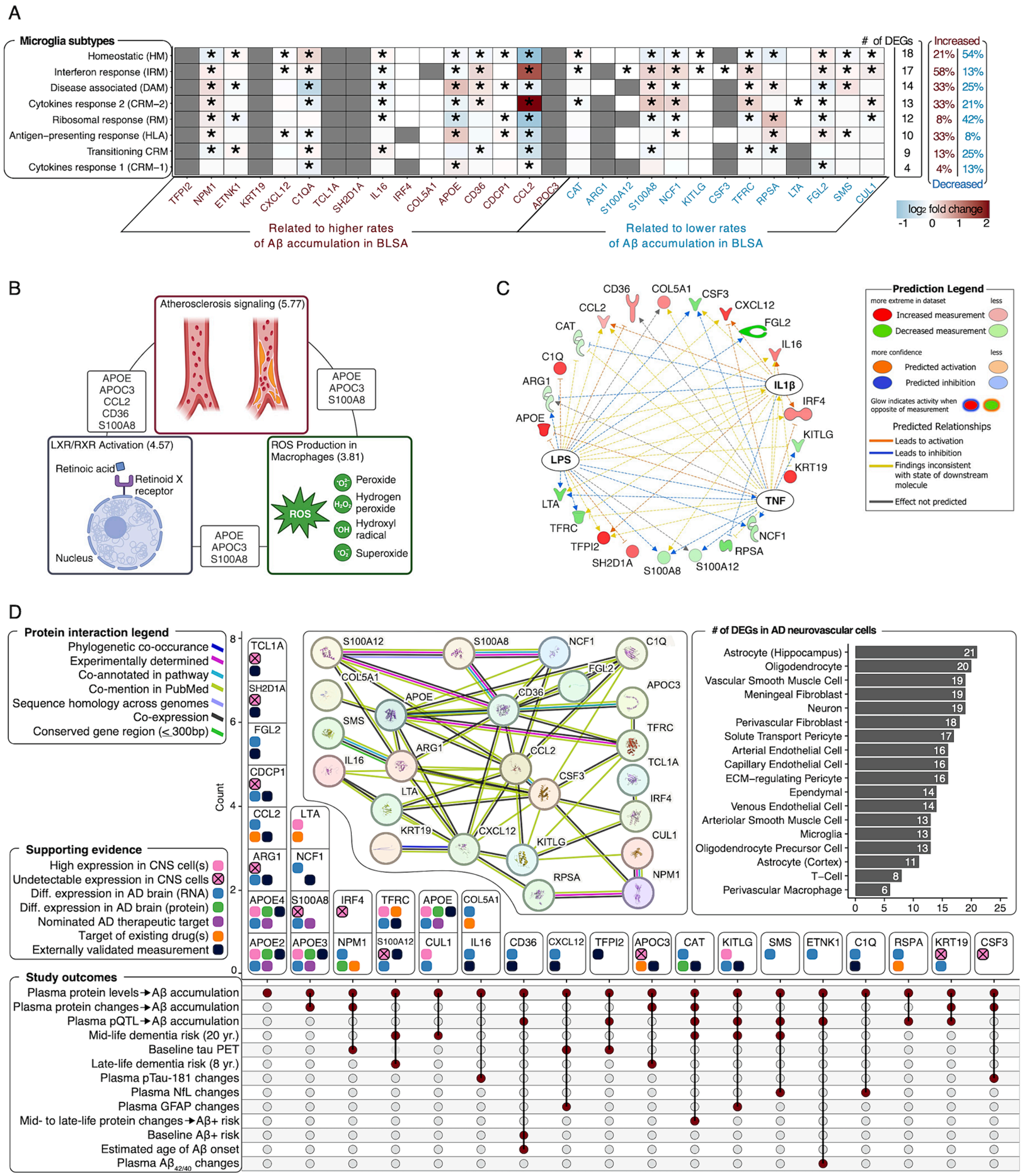

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Fig. 4. Biological characterization of candidate proteins and summary of evidence. A Heatmap shows differential expression of cognate genes encoding candidate proteins across eight distinct microglia subtypes from single-cell transcriptomics of human microglia xenotransplanted to mouse brains. Results derived from Wilcoxon rank sum tests comparing expression levels in a given subtype relative to average expression in other subtypes. *Indicates statistically significant associations (Bonferroni corrected $p < 0.05$). Gray tiles indicate no expression was detected. B An illustrative representation of the top three canonical biological pathways enriched among candidate proteins. Values in parentheses correspond to −log Benjamini-Hochberg corrected *p*-values. Gray text in-between nodes corresponds to overlapping candidate proteins across respective biological pathways. Results derived from Ingenuity Pathway Analysis (IPA) and the illustration was created in BioRender. C A radial plot shows the top three upstream regulators of candidate proteins (TNF, IL1-β, LPS), as well as their predicted effects on specific candidate proteins. Shapes correspond to protein type (, cytokine; , enzyme; , transmembrane; , transcription regulator; , transporter; , peptidase; , translation regulator; , other). Results derived from IPA. D A modified upset plot summarizing evidence for candidate proteins, including empirical results obtained for the current study's outcomes, protein–protein interaction networks (results derived from STRING; predicted protein conformations are depicted in each circular node), a bar graph showing the frequency of cognate genes encoding candidate proteins differentially expressed among neurovascular cell types in AD brain tissue (results derived from Welch's *t*-test using data from the Human BBB), and supporting evidence in the form of CNS cell specific expression patterns (using data from the Human Protein Atlas), expression levels (RNA, protein) in post-mortem AD brain tissue and nominated therapeutic targets (using data from the Accelerating Medicine Partnership for AD), targets of known medications (using data from the Open Targets platform), and externally validated measurements using one or more biochemical techniques (e.g., mass spectroscopy, proximity extension assay etc.,). Key: Aβ, amyloid beta; AD, Alzheimer's disease; CNS, central nervous system; DEG, differentially expressed gene; IL1β, interleukin-1-beta, LPS, lipopolysaccharide; ROS, reactive oxygen species; TNF, tumor necrosis factor.

variety of complementary biological mechanisms, including recruiting peripheral immune cells, modulating proteostasis, and regulating BBB permeability (sTable 30; Fig. 5). Protein interaction analyses revealed an extensive network among candidate proteins, including ample co-expression patterns (Fig. 4D; sTable 31). Lastly, we found 15 medications that target the levels of seven candidate proteins, suggesting that such drugs may have potential for repurposing in the treatment of AD, especially among individuals vulnerable to accelerated cortical Aβ accumulation (Fig. 4D; sTable 32).

3. Discussion

Using data from the large, community based BLSA cohort, we identified 32 immunological proteins in plasma, including CAT, CD36, and KRT19, that were associated with rates of cortical Aβ as measured by Aβ PET (average follow up: 5 years/3.8 scans), with longitudinal changes in a subset of proteins also linked to cortical Aβ progression. After demonstrating that changes in plasma levels of one candidate protein, CAT, between mid- and late-life were associated with elevated Aβ PET in the ARIC cohort, we identified genetic variants associated with plasma levels of CAT, CD36, and KRT19 that also predicted rates of Aβ accumulation in the BLSA, suggesting a potential mechanistic role of these proteins in Aβ progression. Additionally, we demonstrated that several candidate proteins were associated with other AD phenotypes in the BLSA, including tau PET and longitudinal plasma ADRD biomarker changes, and that 31 % of candidate proteins were related to mid-life (20-year) and/or late-life (8-year) dementia risk in ARIC. Finally, we showed the biological implications and functional relevance of candidate proteins, including evidence of prominent downregulation in HM and upregulation in IRM and DAM human microglia subtypes, as well as differential expression patterns across neurovascular cell types in AD brain tissue. In addition to identifying plasma proteins associated with longitudinal Aβ accumulation, we nominate specific immune mediators that might contribute to rates of this pathophysiological process in AD.

The current investigation extends our understanding of how the peripheral immune proteome relates to *in vivo* Aβ pathology beyond

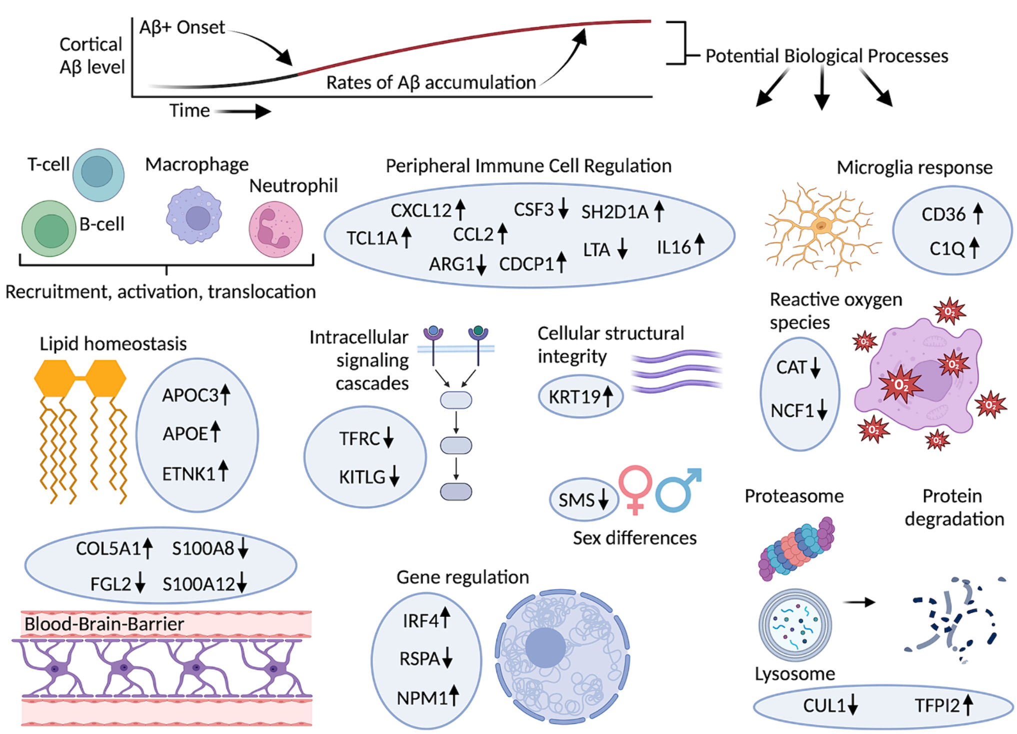

Fig. 5. Potential roles of candidate proteins in regulating Aβ progression. Arrows correspond to the direction of associations between candidate proteins and rates of Aβ accumulation in the BLSA. Canonical functions of individual proteins were obtained from GeneCards (https://www.genecards.org) and the Human Protein Atlas (https://www.proteinatlas.org).

select inflammatory markers measured in single-plex assays. Higher CRP in blood has been linked to increased Aβ burden up to two decades later (Walker et al., 2018), and increased plasma IL-6 and CRP have been related to accelerated Aβ deposition among Aβ+ individuals over a shorter follow-up time (2-years) (Oberlin et al., 2021). Although IL-6 and CRP did not emerge as candidate proteins in our analyses, we suspect that the patterns of protein expression reported here reflect brain Aβ levels because i) the majority of candidate protein measurements have been externally validated, ii) every cognate gene with available expression data was differentially expressed in AD brains, and iii) the majority of cognate genes were differentially expressed in DAM and IRM, well-characterized microglia phenotypes exhibited in response to Aβ (Deczkowska et al., 2018; Sala Frigerio et al., 2019). Almost all proteins associated with Aβ progression were unrelated to Aβ status, suggesting that the peripheral immune processes implicated in initial Aβ deposition may be distinct from the mechanisms subsequently involved in progressive Aβ accumulation, a postulation supported by the limited predictive validity of AD-polygenic risk scores for Aβ PET status compared to clinical diagnosis (Altmann et al., 2020; Leonenko et al., 2019). The counterintuitive associations of APOE proteins (i.e., APOE protein levels are lower with *APOE*4 genotype and in AD (Giannisis et al., 2022)) with faster Aβ accumulation (but not Aβ status) additionally supports this postulation, in light of accumulating data suggesting that initial Aβ deposition is almost entirely *APOE* dependent, whereas Aβ accumulation and disease progression in response to initial Aβ deposition is likely accounted for by non-*APOE* mechanisms (Leonenko et al., 2019; Suzuki et al., 2020; Evans et al., 2023). Our study's unique integration of longitudinal PET imaging with longitudinal proteomic measurements adds to prior evidence showing that both levels and rates of change in circulating immune proteins may drive parallel rates of change in Aβ pathology[3], and indicates that cortical Aβ trajectories may be accompanied by gradual increases and decreases in protective and pathogenic proteins, respectively.

Our finding that increased levels of some circulating immune proteins can be related to both beneficial and deleterious effects downstream of Aβ deposition (i.e., changes in ADRD plasma biomarkers and dementia risk) aligns with prior evidence indicating that such relationships can be dependent on disease stage (Oberlin et al., 2021; Yang et al., 2022; Pereira et al., 2022; Brosseron et al., 2022). This interpretation is supported by i) our primary analyses, where proteins related to greater Aβ progression among Aβ+ participants tended to be associated with lower Aβ progression among Aβ− participants, ii) our dementia-risk analyses, where TFRC was related to lower and higher risk depending on if it was measured at mid-or late-life, along with several other proteins showing similar (albeit statistically insignificant) transitions across this 20 year interval, and iii) our examination of single cell RNAseq data in AD brain tissue, where patterns of differential expression were more apparent in later stages of AD pathology relative to earlier stages. In addition, previously described pleiotropic relationships (Ghosh et al., 2013; Lee et al., 2010) between inflammatory markers and distinct aspects of AD pathology may account for some of the unexpected associations we observed (e.g., IL16 and COL5A1 predicted higher Aβ slopes and lower pTau-181 increases).

Although we did not detect robust expression of cognate genes across total microglia cell populations, cognate genes encoding candidate proteins tended to be downregulated in a human microglia subtype that is responsible for maintaining cellular homeostasis (HM) and upregulated in human microglia subtypes that are exhibited in response to Aβ (DAM, IRM). These findings suggest a transition of microglia expression patterns from homeostatic to Aβ−responsive in the context of Aβ plaque accumulation may partially contribute to the observed associations between candidate proteins and cortical Aβ trajectories. The majority of candidate proteins associated with Aβ accumulation were not highly expressed in CNS cells under normal physiological conditions, and only two showed associations with a biomarker of astrocyte-mediated neuroinflammation (GFAP), but the consistent detection of their differing RNA (and even some protein) concentrations in AD brains suggests that proteins emanating from peripheral immune cells may be particularly important drivers of Aβ neuropathology, potentially through interactions with neurovascular and meningeal cell types (Yang et al., 2021; Da Mesquita et al., 2018). This interpretation is supported by our finding that candidate proteins were enriched for atherogenic signaling pathways, susceptible to regulation by key pro-inflammatory cytokines (e.g., TNF, IL1-β), and differentially expressed across many neurovascular cell types in AD.

Genetic variants modulating plasma levels of CD36, KRT19, and CAT were associated with rates of Aβ progression, suggesting these immune proteins may play a mechanistic role in the accumulation of cortical Aβ. CD36, a lipid-regulating scavenger receptor with an established role in cardiovascular disease (Shu, 2022), can mediate microglial responses to Aβ and is genetically linked to age of AD onset (Šerý et al., 2022; El Khoury et al., 2003), while its rs977371848 pQTL (aka rs2519093) is a strong pQTL for CSF levels of FAM3D and SELE, which are themselves implicated in AD (Yang et al., 2021; Hasegawa et al., 2014; Li et al., 2015). KRT19, a cytoskeletal intermediate filament protein, is differentially expressed across the AD spectrum in patient-derived stem cells (Rantanen et al., 2022), and its rs114115200 pQTL is strongly associated with plasma levels of EHMT2, whose *in vivo* modulation can rescue synaptic and cognitive deficits in AD mice (Yang et al., 2021; Zheng et al., 2019). Given the *trans* location of CD36 and KRT19 pQTLs, they may account for observed effects of their respective proteins on cortical Aβ by regulating the levels and/or interactions of their proteins' binding partners, modifying factors upstream or downstream in their proteins' signaling pathways, or distally regulating their proteins' expression through unknown intermediate genes (e.g., transcription factors near these pQTLs which then influence expression of plasma proteins); however, these variants may also affect Aβ accumulation and protein levels through independent mechanisms, suggesting that the observed associations could also reflect pleiotropy. CAT is a neuroprotective antioxidant that can inhibit Aβ−induced neuropathology in preclinical models (Habib et al., 2010; Mao et al., 2012), and its rs1001179 pQTL is a risk-factor for certain types of cancer (Galasso et al., 2022). CAT's promoter-specific *cis* pQTL (specifically the C allele) was associated with faster Aβ progression and higher CAT protein levels in plasma. In ARIC, higher CAT protein levels increased 20-year dementia risk, and increases in CAT protein levels over this same time period increased risk of Aβ+ . In the BLSA, higher CAT protein levels predicted accelerated Aβ accumulation among Aβ− participants, but an inverse association was found among Aβ+ participants, along with greater ITG tau-PET levels. This pattern of results could indicate that upregulated CAT expression is a maladaptive process during prodromal disease stages which is mitigated among carriers of CAT's rs1001179 T allele, who show attenuated accumulation of Aβ in the context of high Aβ burden due to unknown, CAT-independent mechanisms. Alternatively, these results could suggest that upregulated CAT expression is an adaptive process triggered by early brain amyloidosis which is blunted among carriers of CAT's rs1001179 T allele, and ultimately limits further accumulation of Aβ pathology in the context of high Aβ burden, a disease stage otherwise associated with greater tau burden and dementia risk (sFig. 18). Such dual, disease stage-dependent roles of specific immune proteins have been reported elsewhere[3]. Together, these findings indicate that genetic variation modulates the levels of several candidate proteins in plasma, and that such modulation may be mechanistically relevant to rates of cortical Aβ accumulation.

While our study has several strengths, including the use of longitudinal PET imaging, state-of-the-art quantification of over 900 immunological proteins, single-cell transcriptomics and the assessment of both mid- and late-life dementia risk in an independent cohort, it has several limitations. First, we lacked intrathecal measurements that may have more accurately captured the temporal relationship between protein levels and rates of Aβ progression. Second, our study's modest sample size (particularly Aβ+ participants) likely limited our ability to find associations that might have otherwise been detected in a larger cohort, and to appropriately control for multiple comparisons. Although we leveraged genetics and external validation to reduce potential Type-1 error, and despite many candidate proteins have been previously implicated in AD (e.g., C1Q, APOE[10]), future studies will be needed to validate their associations with altered Aβ trajectories. Third, although sex was included as a covariate throughout analyses, we cannot rule out the possibility that the observed associations may have been modified by variation in sex hormones (e.g., by modulating protein synthesis (Henderson et al., 2009) and/or Aβ pathology (Lee et al., 2017)). Fourth, because candidate proteins showed minimal overlap with proteins related to Aβ positivity, their clinical utility as standalone biomarkers is likely limited, especially during pre-clinical disease stages. Fifth, although the majority of cognate genes were differentially expressed in multiple cell types from AD brain tissue and across different microglia subtypes, the statistical significance of these associations should be interpreted with caution as such single-cell RNAseq experiments can be confounded by inflated p values and increased Type-1 error (Squair et al., 2021). Despite these limitations, our results show a link between peripheral immune proteins, rates of cortical Aβ accumulation, and long-term neurocognitive outcomes, while highlighting the immunological proteins in plasma that may influence rates of AD pathophysiology.

4. Methods

4.1. Study sample

The current study used data from the Baltimore Longitudinal Study of Aging (BLSA), an ongoing longitudinal study (started in 1958) designed to assess physical and psychological aging in a cohort of community-dwelling volunteers (Shock et al., 1984). Participants received comprehensive health and functional screening evaluations at each study visit (including blood draws), which were completed by licensed health-care professionals (e.g., nurse practitioner, medical doctor). Study visits occurred biennially until 2005, then every 1 to 4 years depending on age (age $<$ 60 years, every 4 years; age 60–79 years, every 2 years; age $\geq$ 80 years, every year). Plasma was collected using standardized protocols and frozen at −80 °C until analysis. PET scans for ^{11}C-Pittsburgh compound-B (PiB) and ^{18}F-flortaucipir (FTP) were initiated in 2005 and 2016, respectively. Due to BLSA's continuous enrollment, participants entered the study at different times and thus varied with respect to follow-up times. Associations of SomaScan proteomic measurements with neurocognitive outcomes used blood samples collected at the baseline visit of each respective outcome (e.g., PET scan). For a subset of participants without a blood sample available at baseline, a blood sample from the next closest preceding BLSA visit was used (sTable 2); participants were excluded if the closest preceding visit with an available blood sample was $>$ 1 visit before baseline.

Participants were also excluded for missing data, neurovascular conditions that could affect brain structure or function (e.g., strokes), and baseline cognitive impairment (i.e., dementia, MCI, impaired but not MCI; procedures for determining cognitive status are described elsewhere (Kawas et al., 2000). Reporting follows Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) reporting guidelines.

4.2. Positron emission tomography (PET)

^{11}C-Pittsburgh compound-B (PiB) distribution volume ratios (DVR) and ^{18}F-flortaucipir (FTP) standardized uptake value ratios (SUVR) were measured using positron emission tomography (PET) (Bilgel et al., 2023). Aβ PET scans (70 min.) were acquired on a GE Advance or Siemens High Resolution Research Tomograph (HRRT) scanner immediately following an i.v. bolus injection of approximately 555 MBq of the radiotracer. DVRs were computed with a spatially constrained simplified reference tissue model using cerebellar gray matter as a reference. Mean cortical Aβ reflected the average DVR values across the cingulate, frontal, parietal (including precuneus), lateral temporal, and lateral occipital regions, excluding the pre- and post-central gyri. Mean cortical DVR values were harmonized between the two scanners by leveraging longitudinal data available on both scanners for 79 participants. Aβ PET status (+/−) was defined based on a Gaussian mixture model threshold of 1.064 mean cortical DVR (Bilgel et al., 2023). Age of Aβ onset was calculated as described previously (Bilgel et al., 2016) with non-linear mixed-effects models incorporating random effects and reflected the estimated age at which a participant became Aβ+. Participants who converted from Aβ− to Aβ+ (n = 7) had $\geq$ 3 scans; for these participants, observations at which a participant's Aβ status reflected the participant's status at $\geq$ 50 % of scans were preserved. One participant who converted from Aβ+ to Aβ− across their only two scans was excluded. Tau PET scans (30 min.) were acquired on a Siemens HRRT scanner 75 min. after an i.v. bolus injection of approximately 370 MBq of the radiotracer. Using PET images partially volume corrected with a region-based voxel-wise method, SUVRs were computed with the inferior cerebellar grey matter as a reference. Averaged bilateral SUVRs for five regions of interest- Braak stages (I-II, II-IV, and V-VI), the entorhinal cortex (ETC) and the inferior temporal gyrus (ITG)- were examined due to their relevance to cortical Aβ pathology (Schöll et al., 2016; Bilgel et al., 2022). Braak stage specific SUVR calculations utilized an established Braak staging system that approximated the anatomical definitions of transentorhinal (I-II), limbic (III-IV) and isocortical (V/VI) Braak stages (Schöll et al., 2016).

4.3. Plasma protein measurements

Immune proteins in plasma were measured with the SOMAmer-based array method (SomaLogic) as described previously (Candia et al., 2022), with analyses restricted to the Inflammation and Immune Response Panel that initially contained 938 SOMAmer reagents (aptamers) from the larger SomaScan v4.1. Samples from participants that did not pass SomaScan quality control (QC) criteria were excluded (n = 7). An additional 2 aptamers were included in analyses to capture variation in APOE3 and APOE4, and four other aptamers were included to capture variation in two proteins (CDCP1, ITGA11) that we recently showed to be associated with inflammatory diet and cognitive impairment (Duggan et al., 2023). Samples identified as outliers by SomaScan or by principal component analyses were excluded. Using 102 blind duplicates, aptamers with intra-assay coefficients of variation (CV) $>$ 50 % were excluded (n = 2). The median CV for aptamers on the Inflammation and Immune Response Panel was 4.5 %. Values were $\log_2$ transformed and those beyond 5 SDs were winsorized. EntrezGeneIDs were used as labels for corresponding aptamers.

AD and related dementia (ADRD) plasma biomarkers were measured on the Single Molecule Array (Simoa) HD-X instrument (Quanterix). $A\beta_{40}$, $A\beta_{42}$, GFAP, NfL were quantified using Neurology 4-Plex E (N4PE) assay; CVs were 1.5, 1.0, 4.9, and 4.8 % for $A\beta_{40}$, $A\beta_{42}$, GFAP, NfL, respectively. In participants without PET scans (and a subset of participants with PET scans, n = 97), pTau-181 measurements used a standard Simoa assay (V2); the CV was 4.4 %. Such assays were run in duplicate, and values averaged. Values for GFAP, NfL and pTau-181 were $\log_2$ transformed to correct for skewness. Aβ plasma status (+/−) was defined based on a threshold of 0.0462 plasma $A\beta_{42/40}$ levels. $A\beta_{42/40}$ was used to define Aβ status rather than pTau-181 or $A\beta_{42}$/pTau-181 so that such cut-points could be applicable to the larger set of participants with concurrent SomaScan measurements (i.e., n = 1,015 with N4PE measurements versus n = 688 with V2 measurements) and because its optimal cut point illustrated comparatively similar performance in discriminating Aβ PET status (sFig. 11B-C). Cut points for each biomarker group reflected thresholds that predicted Aβ PET status with a maximized Youden index. Cut point calculation used the subset of participants with concurrent PET, N4PE, and V2 measurements (n =

97), rather than in-house pTau-181 and pTau-231 assays (see below for further description), so that such cut-points could be applicable to the larger set of participants with concurrent SomaScan and V2 measurements (i.e., n = 688 with V2 measurements versus n = 196 with in-house pTau measurements). To directly compare the strength of associations between SomaScan proteins and ADRD biomarkers with respect to rates of cortical Aβ change, pTau-181 and pTau-231 Simoa assays developed and validated in-house at the University of Gothenburg (Karikari et al., 2020) were used because they were available for all Aβ Pet participants (n = 196). For these pTau measurements, concentrations below limit of detection were imputed at 0 and values below lower limit of quantitation were retained as is; repeatability coefficients were 5.3 % for pTau-181 (5.1 % for 11.6 pg/mL and 5.5 % for 15.5 pg/mL) and 5.8 % for pTau-231 (3.4 % for 31.6 pg/mL and 7.4 % for 42.7 pg/mL). Biomarker values were standardized and those beyond 5SDs were winsorized. When directly comparing the strength of associations between SomaScan proteins and ADRD biomarkers with respect to rates of cortical Aβ change, measurements were retained on a $\log_2$ scale to facilitate direct comparison of effect sizes.

4.4. Protein quantitative train loci (pQTLs) and two-sample MR

pQTLs were obtained from deCODE genetics, which conducted a GWAS of SomaScan plasma protein levels (Ferkingstad et al., 2021). For our analyses, we incorporated pQTLs that met a genome-wide significance *p*-value of 1.8 x 10^{-9}. Genome-wide genotyping in the BLSA was performed using the Illumina 550 K or NeuroChip platforms using standard QC procedures described previously (Armstrong et al., 2020). Variants were excluded for poor call rate (missing > 1 %), violations of Hardy-Weinberg Equilibrium ($p < 1 \times 10^{-6}$) and limited minor allele frequency (<1%). Samples were excluded for poor genotyping efficiency (missing > 2 %), sex inconsistencies (i.e., discordance between chromosomal and self-reported sex), cryptic relatedness (pihat > 0.25), or if they were not of European ancestry by self-report or principal component detection. Imputation was performed for Illumina 550 K or NeuroChip datasets separately with the Michigan Imputation Server MaCH (https://imputationserver.sph.umich.edu/) using the HRC r1.1.2016 reference panel. After imputation, datasets were merged, overlapping samples were removed, and SNPs with low imputation quality (R^2 < 0.9), minor allele frequencies < 1 %, Hardy-Weinberg Equilibrium *p*-values $< 1 \times 10^{-6}$, and those that did not overlap between the 2 datasets (missingness < 99 %) were excluded. The final genetic dataset included 5,439,477 SNPs. For two-sample MR, we used deCODE genome-wide significant plasma pQTLs as genetic instruments for candidate plasma proteins. The outcome (Aβ PET levels) was characterized using summary statistics from a GWAS of Aβ PET levels (n = 13,409) (Ali et al., 2023). Plasma pQTLs were pruned to remove genetic variants in linkage disequilibrium ($r^2 < 0.05$) with the 1000 Genomes Project European as the reference panel. The random-effects inverse variance weighted estimate (for proteins with multiple genetic instruments) or the Wald ratio estimate (for proteins with a single genetic instrumental variable) were used for primary analyses. Sensitivity analyses were performed to assess the violation of MR assumptions using MR-Egger, weighted-median, and weighted modes. Assuming sufficient numbers of available instruments, heterogeneity between causal estimates was tested using Cochran's Q and/or the Mendelian Randomization Pleiotropy Residual Sum and Outlier (MR-PRESSO) Global test, and horizontal pleiotropy assumptions were tested using the MR-Egger intercept.

4.5. Covariates

Baseline age (years), sex (male/female), race (white/non-white), and education level (years) were defined based on participant self-reports. *APOEε4* carrier status (0 ε4 alleles/≥1 ε4 alleles/missing) was defined via PCR with restriction isotyping using the Type IIP enzyme Hhai or the Taqman method. Estimated glomerular filtration rate (eGFR)-creatinine was defined at the time of blood sample collection using the CKD-EPI criteria. Comorbid diseases which represent potential confounders were defined using a comorbidity index calculated as the sum (score range: 0–8; converted to a percentage to account for missing data) of eight conditions: obesity, hypertension, diabetes, cancer, ischemic heart disease, chronic heart failure, chronic kidney disease and chronic obstructive pulmonary disease (Duggan et al., 2022).

4.6. Atherosclerosis Risk in Communities (ARIC) study

ARIC is a prospective epidemiologic study conducted in four U.S. communities: Forsyth County, NC; Jackson, MS; the northwest suburbs of Minneapolis, MN; and Washington County, MD. The ARIC study enrolled 15,792 mostly white and black participants aged 45–64 between 1987 and 1989 (Wright et al., 2021). After initial enrollment, participants had four additional in-person visits: Visit 2 (1990–1992), Visit 3 (1993–1995), Visit 4 (1996–1999), Visit 5 (2011–2013). Participants were invited back for Visit 6 (2016–2017) after five years, and Visit 7 (2018–2019) immediately thereafter. Blood was drawn for proteomic analysis at Visits 2 and 5. Aβ PET scans were acquired at ARIC Visit 5. Mid-life (20-year) dementia risk was assessed between Visit 2 and 5, and late-life (8-year) dementia risk was assessed between Visit 5 and 7.

Proteins were measured in ARIC using the SomaScan v4.0 assay (SomaLogic). Protein QC steps for protein measurements in ARIC have been described previously (Walker et al., 2021). Using 197 blind duplicates, the median CV for aptamers examined in the current analyses was 12.4 % and 7.9 % at Visit 2 and 5, respectively. Values were $\log_2$ transformed and those beyond 5 SDs were winsorized. ^{18}F-florbetapir PET scans (20 min.) were acquired on Siemens scanners (various models across ARIC sites) 50–70 min. after an i.v. bolus injection of the radiotracer, as described previously (Gottesman et al., 2016). SUVRs were calculated using cerebellar gray matter region as a reference. Mean cortical Aβ reflected average SUVR values across the orbitofrontal, prefrontal, and superior frontal cortices, lateral temporal, parietal, and occipital lobes, precuneus, and anterior and posterior cingulate regions of interest. Aβ PET status (+/−) was defined based on a median of 1.20 mean cortical SUVR. Dementia diagnosis was adjudicated through cognitive assessment tests, telephone screening, informant ratings, hospital records, and death record review, as previously described (Knopman et al., 2016). At ARIC Visits 2–4, a 3-instrument cognitive assessment was applied (delayed word recall task, digit symbol substitution from the Wechsler Adult Intelligence Scale-Revised (WAIS-R), and a letter fluency task). At ARIC Visits 5–7 dementia was classified using a surveillance approach, where participants received a comprehensive cognitive exam and a functional assessment that included the Clinical Dementia Rating Scale (CDR) and Functional Activities Questionnaire (FAQ). Using this data, dementia was classified based on the NIA/AA (National Institute on Aging and Alzheimer's Association) and the Diagnostic and Statistical Manual of Mental Disorder – Fifth Edition (DSM-5) criteria. In the time between study visits, participants were contacted annually via phone and administered the SIS, a brief cognitive assessment. If participants received a low score on the SIS, or if they were unable to participate in the screening via phone, the Ascertain Dementia 8 (AD8) was administered to the participant's informant. For participants who received a dementia diagnosis at Visit 6 or 7, SIS, AD8, hospital discharge, and death certificate codes were used to define the date of dementia onset. For participants who did not attend Visits 6 or 7, SIS, AD8, hospital discharge, and death certificate codes were used to define dementia diagnosis and date of dementia onset.

4.7. Johns Hopkins Neurology Clinic

CSF and plasma samples from the same participants were used for proteomic analyses using SomaScan v4.1 assay. Samples that did not pass SomaScan QC criteria were excluded (CSF n = 0; plasma n = 2).

Samples identified as outliers by SomaScan or by principal component analyses were excluded (CSF n = 3; plasma n = 1). CVs were calculated using QC pooled, matrix-matched sample replicates provided by SomaLogic to monitor overall assay performance. The paired plasma-CSF study design spanned 3 plates for each biological matrix (plasma or CSF) and included 3 matrix-matched QC sample replicates in each plate. CVs were calculated as CV = 100*sd(RFU)/mean(RFU), where RFU is the SOMAmer intensity in Relative Fluorescence Units measured for all intra- and inter-plate QC samples. Aptamers with CVs > 50 % were excluded (CSF n = 15; plasma n = 18). The median CV for candidate protein plasma aptamers used in the current analyses was 5.8 % for CSF and 4.0 % for plasma. Values were $\log_2$ transformed and those beyond 5 SDs were winsorized. EntrezGeneIDs were used as labels for corresponding aptamers.

4.8. Xenotransplanted human microglia

We examined the expression of cognate genes encoding candidate proteins in a single-cell transcriptomic dataset of human microglia xenografted to mouse brains (Mancuso et al., 2024). Briefly, microglia precursors from multiple cell lines (UKBIO11-A, BIONi010-C and H9) were transplanted to the brains of 4 day old neonates (P4) of three different host genetic backgrounds (humanized *App* control mice, *Rag2*$^{-/-}$ *Il2rγ*$^{-/-}$ *hCSF1*^{KI} *App*$^{hu/hu}$; humanized *App* knock-in mice, *Rag2-/- Il2rγ*$^{-/-}$ *hCSF1*^{KI} *App*$^{NL\text{-}G\text{-}F}$; *ApoE* knock-out mice, *Rag2-/- Il2rγ*$^{-/-}$ *hCSF1*^{KI} *App*$^{NL\text{-}G\text{-}F}$ *Apoe*$^{-/-}$) using the MIGRATE protocol and isolated 6–7 months later using fluorescence-activated cell sorting applied to dissociated brain tissue. Transcriptomes from 138,577 microglia cells (n = 106, 20 sequencing libraries; HiSeq4000 or NovaSeq6000 [Illumina]) subjected to rigorous QC procedures were combined and then clustered into distinct groups using Uniform Manifold Approximation and Projection (Mancuso et al., 2024). The Wilcoxon rank sum tests were used to determine if $\log_2$ fold changes in a given microglia subtype were significantly different relative to other subtypes. Bonferroni correction was used to adjust *p*-values for multiple comparisons.

4.9. Protein characterization

Several complementary analytical tools were used to understand the biological relevance of proteins. We identified studies (PubMed IDs) that externally validated SOMAmer reagents using a variety of techniques, including: multiple reaction monitoring (MRM) mass spectroscopy (which determines protein content in a single sample after SOMAmer enrichment) and data dependent analysis (DDA) mass spectroscopy (which determines protein content after SOMAmer enrichment for pull down testing in matrix (e.g., plasma)), protein quantitative train loci (which indicate genetic strategies, such as *cis* pQTL associations, directly link a gene to a protein and a SOMAmer), orthogonal strategies (which refer to variety of methods that used SOMAmer reagent-independent techniques to quantify and confirm protein identity and level of expression, including ELISA, histochemistry, western blot, particle-based immunoassay etc.,), and proximity extension assays (which use antibody pairs conjugated to unique oligonucleotides and qPCR to determine that protein levels in plasma and SOMAmers display Spearman's *rho* correlations of > 0.40). Enriched biological pathways and upstream regulators were identified using Ingenuity Pathway Analysis (Qiagen Inc; version 01-22-01), a bioinformatics application that facilitates the interpretation of '-omics' data using manually curated content available through the Ingenuity Knowledge Base (Krämer et al., 2014). Protein levels reflected baseline concentrations associated with rates of change in cortical Aβ. Benjamini-Hochberg FDR adjusted *p*-values derived from Fisher's exact tests quantified the probability of overlap between candidate proteins and molecules (genes, transcripts, proteins, drugs) known to exist within a specific pathway or process due to random chance. Protein interaction networks were assessed using STRING (Search Tool for the Retrieval of Interacting Genes/Proteins) (https://string-db.org), a database and visualization tool that integrates publicly available sources of information to provide a comprehensive understanding of protein–protein interaction (PPI) networks. PPI *p*-values derived from an explicit null model to account for the non-uniform distribution of the connectivity degrees of network proteins (i.e., a random graph with given degree sequence model) reflected the likelihood of proteins having more interactions among themselves than what would be expected for a random set of proteins of the same size drawn from the genome, suggesting if the proteins are at least partially biologically connected as a group (Franceschini et al., 2013). For comparative gene expression across 76 cell types, we used consensus transcript expression levels (normalized Transcripts per Million; nTPM) from the Human Protein Atlas (https://www.proteinatlas.org/about/). Here, cell specific expression, including in CNS cell types (i.e., astrocytes, excitatory/inhibitory neurons, microglia, oligodendrocytes, etc.,), was identified as the top 5 cell types with the highest expression of a given gene; a gene was considered undetectable in CNS cells if all CNS cell types maintained ≤ 1 nTPM. Supplemental information relevant to AD, including expression levels (RNA, protein) in post-mortem brain tissue, was obtained from the AD Knowledge Portal (https://adknowledgeportal.synapse.org), a platform for accessing data, analyses, and tools generated by the Accelerating Medicines Partnership Program for AD (AMP-AD) and other NIA-supported programs. Expression levels in six prominent CNS cell types (excitatory neurons, inhibitory neurons, oligodendrocytes, astrocytes, oligodendrocyte precursor cells, microglia) were obtained from previously published results derived from 48 participants in the Religious Orders Study/Memory and Aging Project (Mathys et al., 2019). Here, early pathology was defined as some amyloid burden, modest neurofibrillary tangles, and modest cognitive impairment; late pathology was defined as high amyloid burden, high neurofibrillary tangles, high global pathology, and moderate/severe cognitive impairment. Expression levels in neurovasculature cell types were obtained from the Human BBB (https://twc-stanford.shinyapps.io/human_bbb/), a transcriptomic dataset generated using VINE (Vessel Isolation and Nuclei Extraction)-seq (Yang et al., 2022). The Open Targets Platform (https://platform.opentargets.org) was used to identify medications that target specific proteins. As no separate Entrez gene IDs were available for APOE isoform aptamers (APOE, APOE2, APOE3, APOE4), cognate gene identification used one gene ID (APOE); as no Entrez gene ID was available for the C1Q aptamer, cognate gene identification used one related gene ID (C1QA).

4.10. Statistical analyses

Participant-specific rates of change were extracted from linear mixed-effects regression models using random effects of intercept and time with unstructured covariance. Model quality and goodness of fit were assessed using the performance R package (0.11.0.8). After the quality of model fit was confirmed, participant-specific rates of change were standardized. Multiple linear regression models adjusting for baseline age, sex, race, education, *APOE* ε4 status, Aβ status (+/−), sample batch (1/2) and a comorbidity index were used to examine associations of proteins with rates of change in mean cortical Aβ and ADRD biomarkers (also adjusted for eGFR-creatinine). Aβ status was defined using Aβ PET status (+/−) for PET analyses; otherwise, it was defined using Aβ PET status and/or plasma Aβ status (+/−), with the former taking precedent. Minimally adjusted sensitivity analyses corrected for baseline age, sex, *APOE* ε4 status, Aβ status (+/−), sample batch (1/2) and a comorbidity index. To determine if sex modified the associations of plasma proteins, a two-way interaction term (i.e., sex*protein) was incorporated into models. An additional sensitivity analyses was conducted to adjust for sample storage duration. Logistic regression models adjusted for the aforementioned covariates (including age) were used to examine associations of proteins with odds of Aβ PET status. Multiple linear regression models adjusting for covariates used in the pQTL discovery cohort (Ferkingstad et al., 2021) (baseline age and sex) were used to examine associations of SNPs with protein levels. Covariate-adjusted multiple linear regression models were used to examine associations of SNPs with rates of change in cortical Aβ. Protein level changes in ARIC were related to odds of Aβ status using logistic regression models adjusting for age, sex, race-center, education, *APOEε*4, eGFR-creatinine, and cardiovascular risk factors (BMI, diabetes, hypertension, and current smoking status). Proteins in ARIC were related to incident dementia risk using Cox proportional hazards regression models adjusting for similar covariates. Details of statistical analyses (e.g., predictor variables, response variables, covariates etc.,) are summarized in the Supplementary materials (sTable 33). Statistical significance was defined at two-sided $p < 0.05$. Analyses were performed using R (4.2.3).

5. Ethics statement

The BLSA protocol was approved by the Institutional Review Board (IRB) of the National Institute of Environmental Health Science, National Institutes of Health (NIH), and the PET studies were approved by the Johns Hopkins University Medicine IRB. All participants gave written informed consent prior to participation and deidentified data were used for analyses. ARIC study protocols were approved by IRBs at each participating center: University of North Carolina at Chapel Hill, Chapel Hill, NC; Wake Forest University, Winston-Salem, NC; Johns Hopkins University, Baltimore, MD; University of Minnesota, Minneapolis, MN; and University of Mississippi Medical Center, Jackson, MS. All ARIC participants gave written informed consent at each study visit; proxies provided consent for participants who were judged to lack capacity. Subjects referred to the Johns Hopkins Hospital Neurology Clinic consented to banking of residual CSF after clinical testing under an IRB approved protocol. Animal experiments were conducted according to protocols approved by the Ethical Committee of Laboratory Animals at the KU Leuven (ECD project number P177/2017) in accordance with country and European Union guidelines.

6. Code availability

No original code was developed for this work. The statistical code is publicly available through GitHub at /dugganmr/amyloid_progression.

CRediT authorship contribution statement

Michael R. Duggan: Writing – review & editing, Writing – original draft, Formal analysis, Conceptualization. **Gabriela T. Gomez:** Writing – review & editing, Formal analysis. **Cassandra M. Joynes:** Formal analysis, Methodology. **Murat Bilgel:** Writing – review & editing, Data curation. **Jingsha Chen:** Writing – review & editing, Formal analysis. **Nicola Fattorelli:** Writing – review & editing, Formal analysis. **Timothy J. Hohman:** Writing – review & editing, Data curation. **Renzo Mancuso:** Writing – review & editing, Data curation. **Jenifer Cordon:** Writing – review & editing, Data curation. **Tonnar Castellano:** Writing – review & editing, Data curation. **Mary Ellen I. Koran:** Writing – review & editing, Data curation. **Julián Candia:** Writing – review & editing, Data curation. **Alexandria Lewis:** Writing – review & editing, Data curation. **Abhay Moghekar:** Writing – review & editing, Data curation. **Nicholas J. Ashton:** Writing – review & editing, Data curation. **Przemysław R. Kac:** Writing – review & editing, Data curation. **Thomas K. Karikari:** Writing – review & editing, Data curation. **Kaj Blennow:** Writing – review & editing, Data curation. **Henrik Zetterberg:** Writing – review & editing, Funding acquisition, Data curation. **Anna Martinez-Muriana:** Writing – review & editing, Data curation. **Bart De Strooper:** Writing – review & editing, Funding acquisition, Data curation. **Madhav Thambisetty:** Writing – review & editing, Funding acquisition. **Luigi Ferrucci:** Writing – review & editing, Funding acquisition, Data curation. **Rebecca F. Gottesman:** Writing – review & editing, Funding acquisition, Data curation. **Josef Coresh:** Writing – review & editing, Funding acquisition, Data curation. **Susan M. Resnick:** Writing – review & editing, Funding acquisition, Data curation. **Keenan A. Walker:** Writing – review & editing, Supervision, Funding acquisition, Data curation.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: H.Z. has served at scientific advisory boards and/or as a consultant for Abbvie, Acumen, Alector, Alzinova, ALZPath, Annexon, Apellis, Artery Therapeutics, AZTherapies, Cognito Therapeutics, CogRx, Denali, Eisai, Nervgen, Novo Nordisk, Optoceutics, Passage Bio, Pinteon Therapeutics, Prothena, Red Abbey Labs, reMYND, Roche, Samumed, Siemens Healthineers, Triplet Therapeutics, and Wave, has given lectures in symposia sponsored by Alzecure, Biogen, Cellectricon, Fujirebio, Lilly, and Roche, and is a co-founder of Brain Biomarker Solutions in Gothenburg AB (BBS), which is a part of the GU Ventures Incubator Program (outside submitted work). K.B. has served as a consultant and at advisory boards for Acumen, ALZPath, BioArctic, Biogen, Eisai, Lilly, Moleac Pte. Ltd, Novartis, Ono Pharma, Prothena, Roche Diagnostics, and Siemens Healthineers, has served at data monitoring committees for Julius Clinical and Novartis, has given lectures, produced educational materials and participated in educational programs for AC Immune, Biogen, Celdara Medical, Eisai and Roche Diagnostics, and is a co-founder of Brain Biomarker Solutions in Gothenburg AB (BBS), which is a part of the GU Ventures Incubator Program, outside the work presented in this paper. R.M. has scientific collaborations with Alector, Nodthera and Alchemab, and has been consultant for Sanofi. B.D.S. is or has been a consultant for Eli Lilly, Biogen, Janssen Pharmaceutica, Eisai, AbbVie and other companies. B.D.S is also a scientific founder of Augustine Therapeutics and a scientific founder and stockholder of Muna Therapeutics.

Data availability

Data generated in the current study are available in this article (and its Supplementary figures or Supplementary tables), upon reasonable request, or in an online public repository. Anonymized BLSA data not published within this article may be shared upon request from qualified investigators. Researchers who wish to use BLSA data are encouraged to develop a pre-analysis plan that can be submitted for approval (https://blsa.nia.nih.gov/how-apply). ARIC proteomic data is available through the National Heart, Lung, and Blood Institute (NHLBI) Biologic Specimen and Data Repository Information Coordinating Center (https://biolincc.nhlbi.nih.gov/studies/aric/). Additional requests for clinical or proteomic data may be submitted to the ARIC Steering Committee and will be reviewed to ensure that data can be shared without compromising participant confidentiality or breaching intellectual property restrictions. Participant-level demographic, clinical and proteomic data may be partially restricted based on prior participant consent, and data sharing restrictions may also be applied to ensure consistency with confidentiality or privacy laws and considerations (https://sites.cscc.unc.edu/aric/). Human microglia gene expression data are available in the Gene Expression Omnibus database and an online platform (data.bdslab.org/Mancuso2022).

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Appendix A. Supplementary data

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